

Isabella Beltran Shapland

510-990-5598 | isabshapland@gmail.com | [LinkedIn](#) | [GitHub](#)

Mechanistic ML Researcher - Computer Science BS 2026 @ University Of California, Santa Cruz
Focus: RL Alignment, Visual Reasoning & Model Reliability

EDUCATION

University of California Santa Cruz, Computer Science B.S.

Santa Cruz, CA

Relevant Coursework: Advanced AI, Applied ML: Deep Learning, DSA. GPA: 3.77

Sept. 2022 – June 2026

EXPERIENCE

UC SANTA CRUZ

Santa Cruz, CA

First Author Machine Learning Researcher (Preprint available soon)

Jan 2026 – Present

- **Breaking the Loop with Mechanistic Attention Probes In-Reward:** Integrated an anti-repetition-collapse correlated visual-attention signal safely into advantage estimation in Group Relative Policy Optimization. Imperfect internal signals can be utilized in reward as long as they are injected safely, e.g. **image-grounding breaks ties but can never override correctness.**
- **Results: +3.4% in-domain on VLAA-Thinking, +6.5% on MathVision** with the gain mechanistically traced to reduced degenerate repetition-collapse. Annotation-free, Full-finetune; Length, Repetition, Random Controls; 8B VLMs on 2×H200. **Submitted to ACL Rolling Review, Aug 3 as first author**, mentored by UCSC's Dr. Yuyin Zhou and Hardy Chen.

TRAINLOOP (YC W25)

San Francisco, CA

Machine Learning Research Intern ([See full paper for details.](#))

Sept. 2025 – Dec 2026

- **ARC-AGI v2 Explainable Reasoning:** Developed a structured decomposition framework for ARC-AGI v2 visual puzzles, achieving traceable model predictions through interpretable natural language “reasoning booklets”.
- **Frontier Model Dynamics:** Quantified a **45.5% “reasoning-execution gap”** in frontier multi-modal models (GPT-5 class) through rigorous error analysis; identified critical failure modes in visual grounding vs. logical deduction.

INTEL

Santa Clara, CA

Machine Learning Software Engineering Intern ([See full article for details.](#))

May 2025 – Sept. 2025

- **RAG Infrastructure:** Architected a Haystack-based RAG pipeline using LLaMA-3.2 to extract CPU anomalies from large-scale JSON system logs; built a deterministic Python validation framework to eliminate hallucinations.
- **Performance Profiling:** Profiled iGPU-accelerated inference using Ollama and SYCL; authored the first internal benchmarks for local model hosting on integrated GPUs.

ASSOCIATION OF COMPUTING MACHINERY (ACM)

Santa Cruz, CA

Machine Learning Researcher

Sept. 2023 – June 2024

- **Adversarial Attacks Research:** Evaluated LLM resilience (BERT) against “Universal Trigger Tokens” using gradient-based optimization attacks. Created a novel measurement for adversarial robustness using this data.
- **Distributed Infrastructure:** Managed large-scale adversarial simulations using Kubernetes and PyTorch; implemented distributed experiment management to parallelize attack vectors across GPU nodes.

PROJECTS

BrowseIQ (Winner, AgentHacks 2025): Developed an autonomous browser agent using agentic system design and real-time LLM analysis to synthesize browsing behavior into actionable visual dashboards.

Classified: A Privacy Auditor (Hackday UCSC): Built an agentic red-team/blue-team privacy auditor that uses tool-calling LLMs to pivot through layered OSINT data and map a target's exposed footprint onto a live visual canvas.

100-Class Few-Shot Classification: Achieved high accuracy on data-constrained datasets (~10 images/class) using ConvNeXt V2, differential LR unfreezing, and 30-crop TTA.

C++ Emulator (Senior Capstone): Engineered a high-performance C++ emulator with custom memory management and CPU cycle timing to execute hex instruction files and render playable games.

SKILLS

Alignment/RL: Verl, GRPO, RLHF, Reward Modeling, Scalable Oversight, LoRA/PEFT, Adversarial Attacks

Multi-Modal: VQA, Visual Reasoning (ARC-AGI), RAG, Mechanistic Probing

Compute/Infra: PyTorch, Kubernetes, SYCL (Intel GPU Acceleration), Ollama, Haystack, Git, LangChain

Languages: Python, C++, C, Java, Haskell, Assembly